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Glutamate is a neurotransmitter involved in the transmission of nerve impulses from pain receptors to the brain. Ziconotide is a drug that can reduce severe, constant pain. Ziconotide blocks the calcium ion channels at some of the synapses which use glutamate.

The transmission of glutamate at synapses is similar to that of acetylcholine.

Explain how ziconotide reduces severe, constant pain.

[5 marks]

Ziconotide is a polypeptide and acts on synapses in the spinal cord. Scientists investigated the effectiveness of ziconotide in reducing severe, constant pain.

Ziconotide was injected into each patient's cerebrospinal fluid that bathes the brain and spinal cord. Patients recorded the intensity of their pain using a statistically valid scale.

0	2	.	2
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Suggest **two** reasons why the patients had ziconotide injected into their cerebrospinal fluid rather than taking a pill containing the drug.

[2 marks]

1

2

Question 2 continues on the next page

Turn over ►

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Ziconotide was injected at $3 \text{ ng kg}^{-1} \text{ h}^{-1}$ for 8 days into each patient.

$$1 \text{ ng} = 1 \times 10^{-9} \text{ g}$$

Calculate the total mass in grams of ziconotide injected after 8 days into a patient with a body mass of 82 kg

Show your working.

[2 marks]

Answer _____ g

0	2	.	4
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When the patients recorded the intensity of pain, suggest **two** reasons why it was important to use a statistically valid scale.

[2 marks]

1 _____

2 _____
